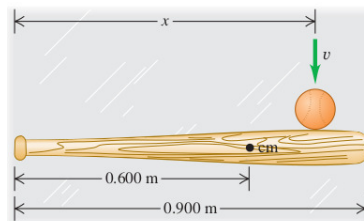


Problem Set 9

Due: 26 July 2024, 3:30 p.m.

Problem 1. A baseball bat rests on a frictionless, horizontal surface. The bat has a length of $l = 0.9$ m, a mass of $m = 0.8$ kg, and its center of mass is $d = 0.6$ m from the handle end of the bat (see figure below). The moment of inertia of the bat about the axis (perpendicular to the surface) through the center of mass is $I_{\text{cm}} = 0.0530 \text{ kg}\cdot\text{m}^2$. The bat is struck by a baseball traveling perpendicular to the bat. The impact applies an impulse of magnitude $J = \int_{t_1}^{t_2} F dt$ at a point a distance x from the handle end of the bat. What must x be so that the handle end of the bat remains at rest as the bat begins to move?

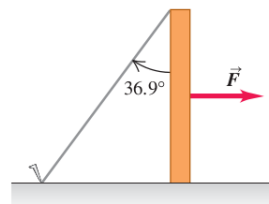


The point on the bat that you will locate is called the *center of percussion*. Hitting a pitched ball at the center of percussion of the bat minimizes the "sting" the batter experiences on the hands.

Hint. Consider the motion of the center of mass and the rotation about the center of mass. Find x so that these two motions combine to give $v = 0$ for the end of the bat just after the collision.

(6 points)

Problem 2. One end of a post weighing 400 N and with height h rests on a rough horizontal surface with $\mu_s = 0.3$. The upper end is held by a rope fastened to the surface and making an angle of 36.9° with the post. A horizontal force $\mathbf{F}$ is exerted on the post as shown. (a) If the force $\mathbf{F}$ is applied at the midpoint of the post, what is the largest value it can have without causing the post to slip? (b) How large can the force be without causing the post to slip if its point of application is $6/10$ of the way from the ground to the top of the post? (c) Show that if the point of application of the force is too high, the post cannot be made to slip, no matter how great the force. Find the critical height for the point of application.



($3/2 + 3/2 + 2$ points)

Problem 3. A circular steel wire of length L must stretch no more than ΔL when a tensile force of magnitude F is applied to each end of the wire. What minimum diameter is required for the wire?

(1 point)

Problem 4. A weight W hangs supported by a wire of length l_0 and cross-sectional area A .

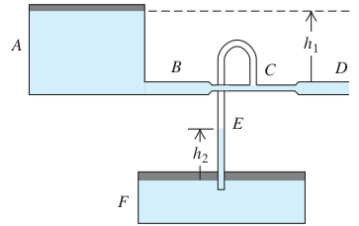
Show that if the wire obeys equation

$$\frac{\text{Stress}}{\text{Strain}} = \text{Elastic modulus},$$

it behaves like a spring of force constant AY/l_0 , where Y is Young's modulus for the material of which the wire is made.

(2 points)

Problem 5. Two very large open tanks A and F contain the same liquid. A horizontal pipe BCD , having a constriction at C and open to the air at D , leads out of the bottom of tank A , and a vertical pipe E opens into the constriction at C and dips into the liquid in tank F . Assume streamline flow and no viscosity. If the cross-sectional area at C is one-half the area at D and if D is a distance h_1 below the level of the liquid in A , to what height h_2 will liquid rise in pipe E ? Express your answer in terms of h_1 .



(4 points)

Problem 6. An open water tank stands on a plane surface. The water surface in the tank is a height h above the plane. A small hole is opened up at a depth y below the surface of the water

- Show that the jet of water will hit the plane surface a distance D from the tank, where $D = \sqrt{4y(h-y)}$.
- Show that the hole should be placed at a depth $y = h/2$ for the jet to cover a maximum horizontal distance.

(2 + 2 points)